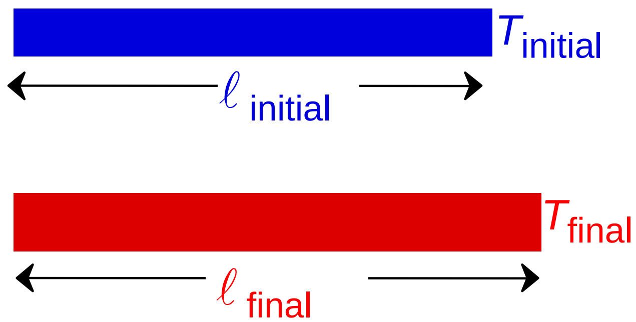
Lecture 23

Thermal Properties 2

Presented by Dr Adijat ‘Wumi Inyang

Thermal expansion

Materials change size with change of temperature



$T_{\text{final}} > T_{\text{initial}}$

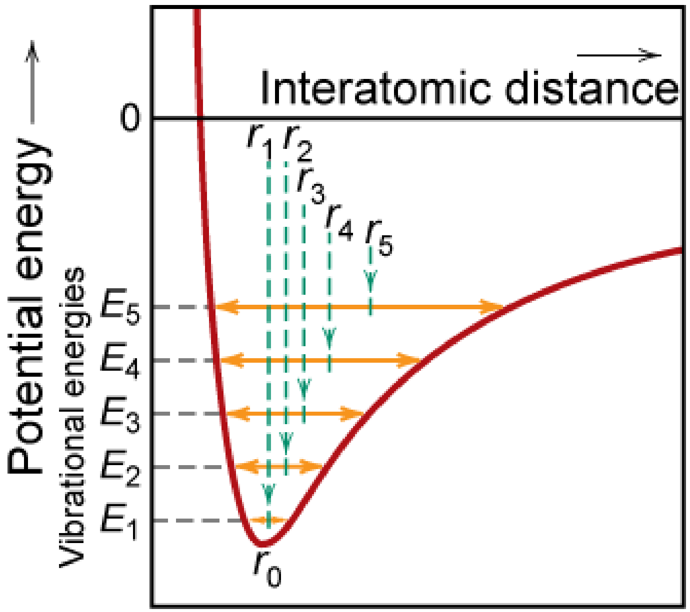
$$\frac{\ell_{\text{final}} - \ell_{\text{initial}}}{\ell_{\text{initial}}} = \alpha_{\ell}(T_{\text{final}} - T_{\text{initial}}) \quad \text{or} \quad \frac{\Delta l}{l_0} = \alpha_l \Delta T$$

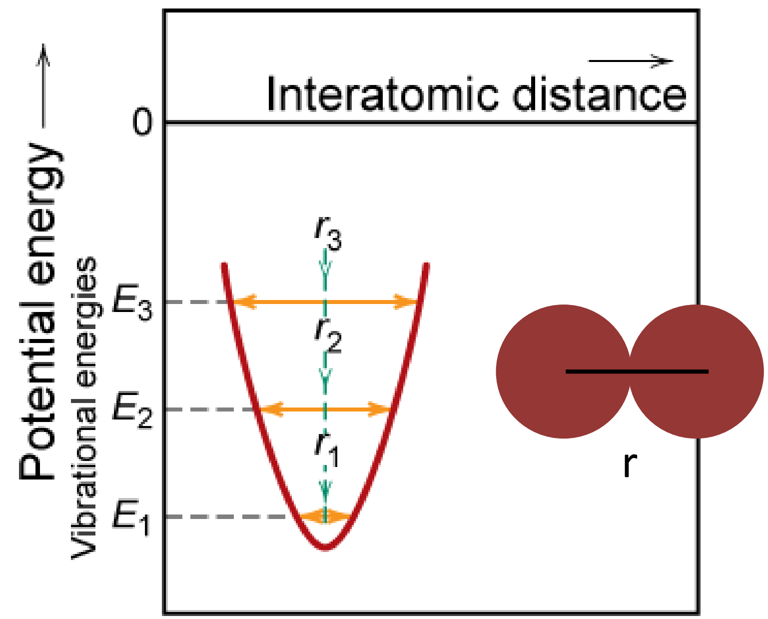
α_l = linear coefficient of thermal expansion (**CTE**)
units of 1/K

Anisotropic materials can have different CTE in different directions

Thermal expansion

Atomic scale: interaction potential between atoms



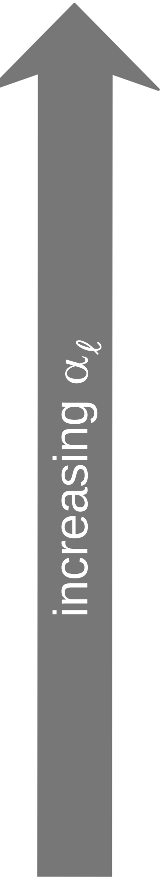


Asymmetric curve:
- increase temperature,
- increase in interatomic separation
- thermal expansion

Symmetric curve:
- increase temperature,
- no increase in interatomic separation
- no thermal expansion

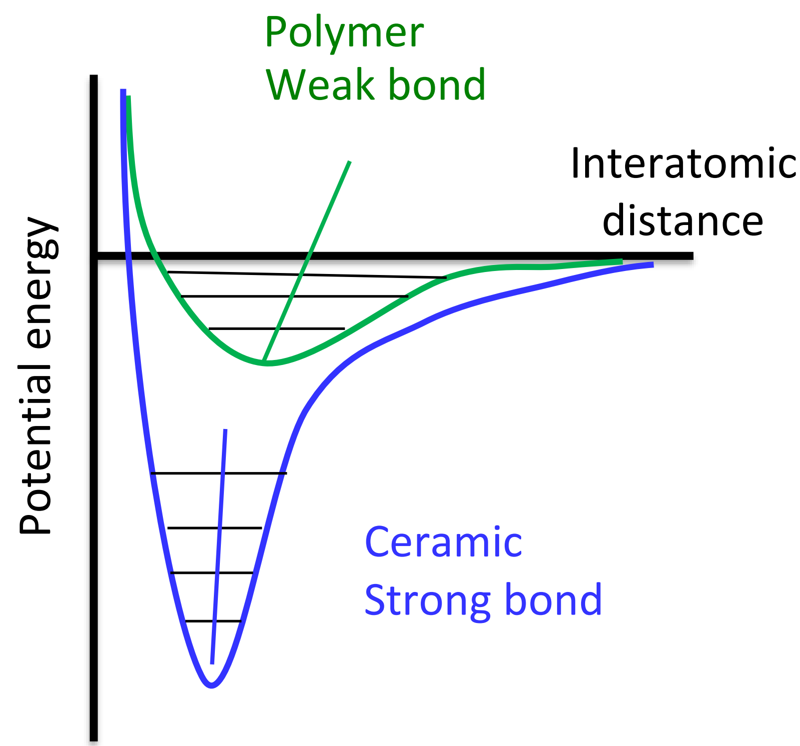
Fig. 20.3, *Callister & Rethwisch 9e*.

Thermal expansion



Material	α_ℓ (10^{-6}/ K) at room T
• Polymers	
Polypropylene	145-180
Polyethylene	106-198
Polystyrene	90-150
Teflon	126-216
• Metals	
Aluminum	23.6
Steel	12
Tungsten	4.5
Gold	14.2
• Ceramics	
Magnesia (MgO)	13.5
Alumina (Al_2O_3)	7.6
Soda-lime glass	9
Silica (cryst. SiO_2)	0.4

Polymers have larger α_ℓ values because of weak secondary bonds



Selected values from Table 20.1, *Callister & Rethwisch 9e*.

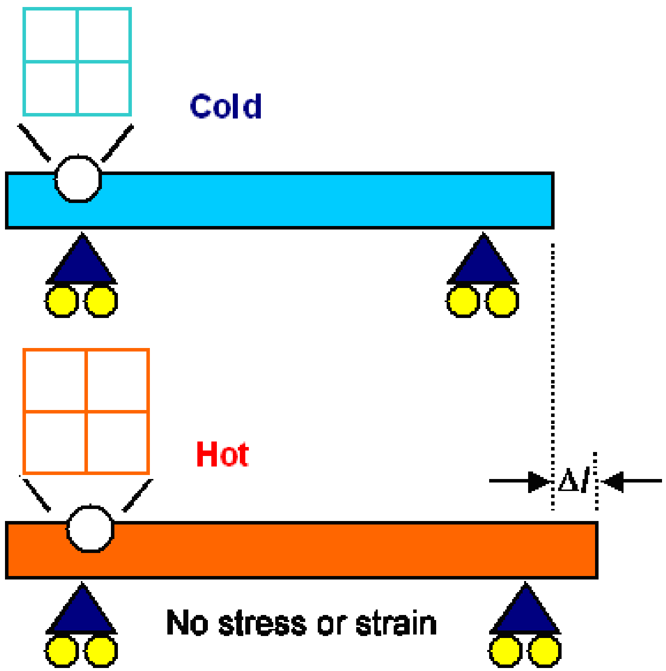
Thermal expansion

Space is left between railway tracks as an allowance for their expansion when temperature increases, otherwise the rails may buckle.

Thermal stresses

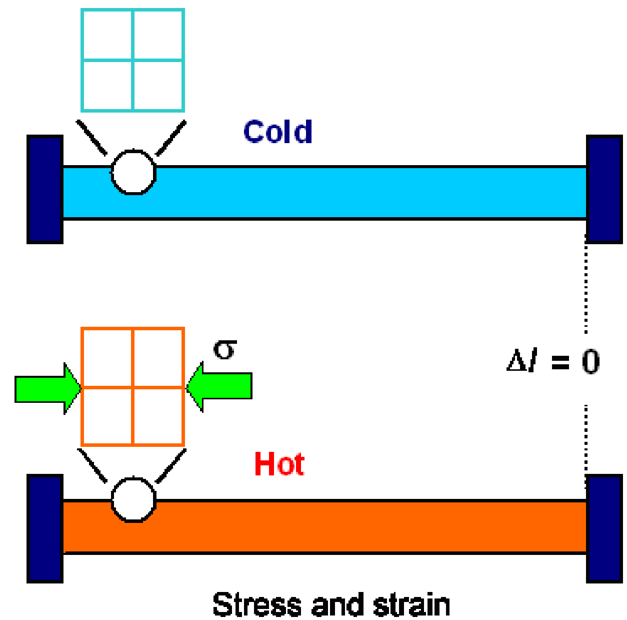
Free to expand

Material extends on heating by Δl



Restrained

Material becomes stressed on heating

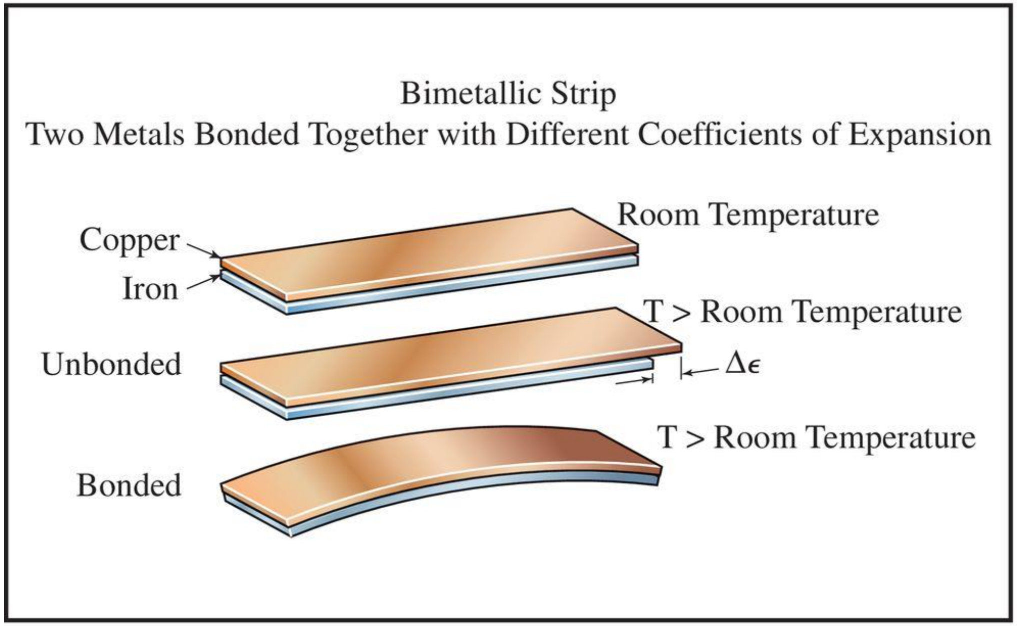


Thermal cycling can lead to thermal fatigue failure

If heating $\sigma < 0$, compressive
If cooled $\sigma > 0$, tension

Differential thermal expansion

Bi-metal strip – 2 metals with different CTE bonded together



Misfit strain $\Delta\varepsilon = (\alpha_1 - \alpha_2)\Delta T$

Radius of curvature, *R*

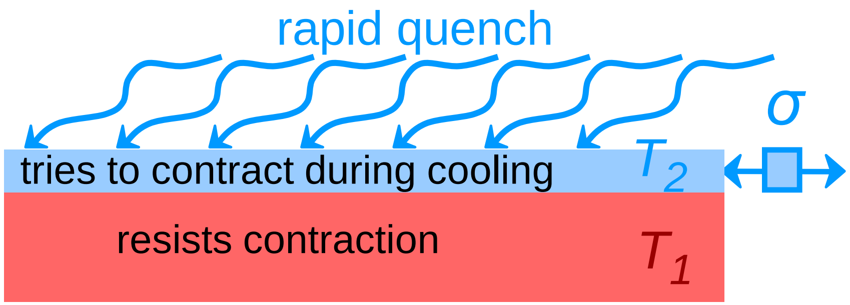
$$\frac{1}{R} = \frac{12\Delta\varepsilon}{h(E_1/E_2 + 14 + E_2/E_1)}$$

where
E = elastic modulus
h = thickness of strip
(same for both metals)
1 and 2 are the two different metals

Used for temperature measurement
Temperature dependent switch (thermostat)
Thermal distortion – change in shape

Thermal shock

Rapid cooling can cause large stresses to be generated



Tension develops at surface

$$\sigma = -E\alpha_{\ell}(T_1 - T_2)$$

Temperature gradient and thermal conductivity combine with thermal expansion to generate large stresses

Tensile stresses at the surface can cause cracks (and fracture, σ_f)

The temperature gradient for failure is given by:

Thermal shock resistance $(TSR) \propto \dfrac{\sigma_f \, k}{E\alpha_{\ell}}$

Thermal shock

Rapid cooling can cause large stresses to be generated in materials with low thermal conductivity

Examples:

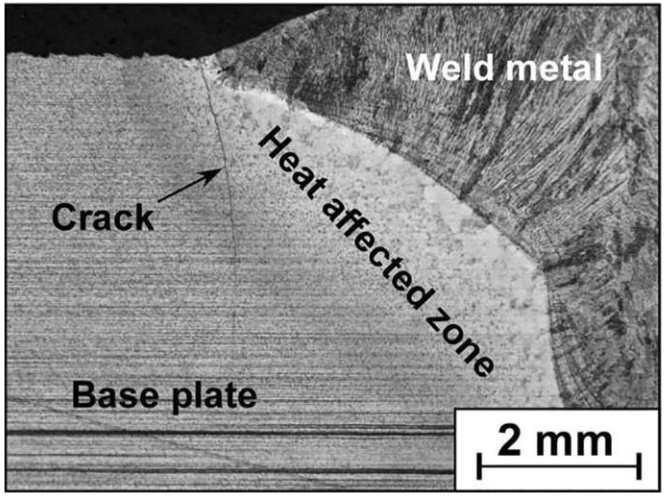
- **Heat treatment of metals**
  – maximum cooling rate (quenching)
  – maximum diameter of metal

$$(TSR) \propto \frac{\sigma_f \, k}{E \alpha_\ell}$$

- **Cooling of ceramics**
  – fracture to make glass powder (frit)

$$(TSR) \propto \frac{K_{IC} \, k}{E \alpha_\ell}$$

- **Welding** in low *k* steels
  – fracture in cooling zone leads to fatigue failure



Applications of thermal properties

- Thermal management

- Photos: Application of ceramic coating Super-therm™

Applications of thermal properties

- Thermal conductivity

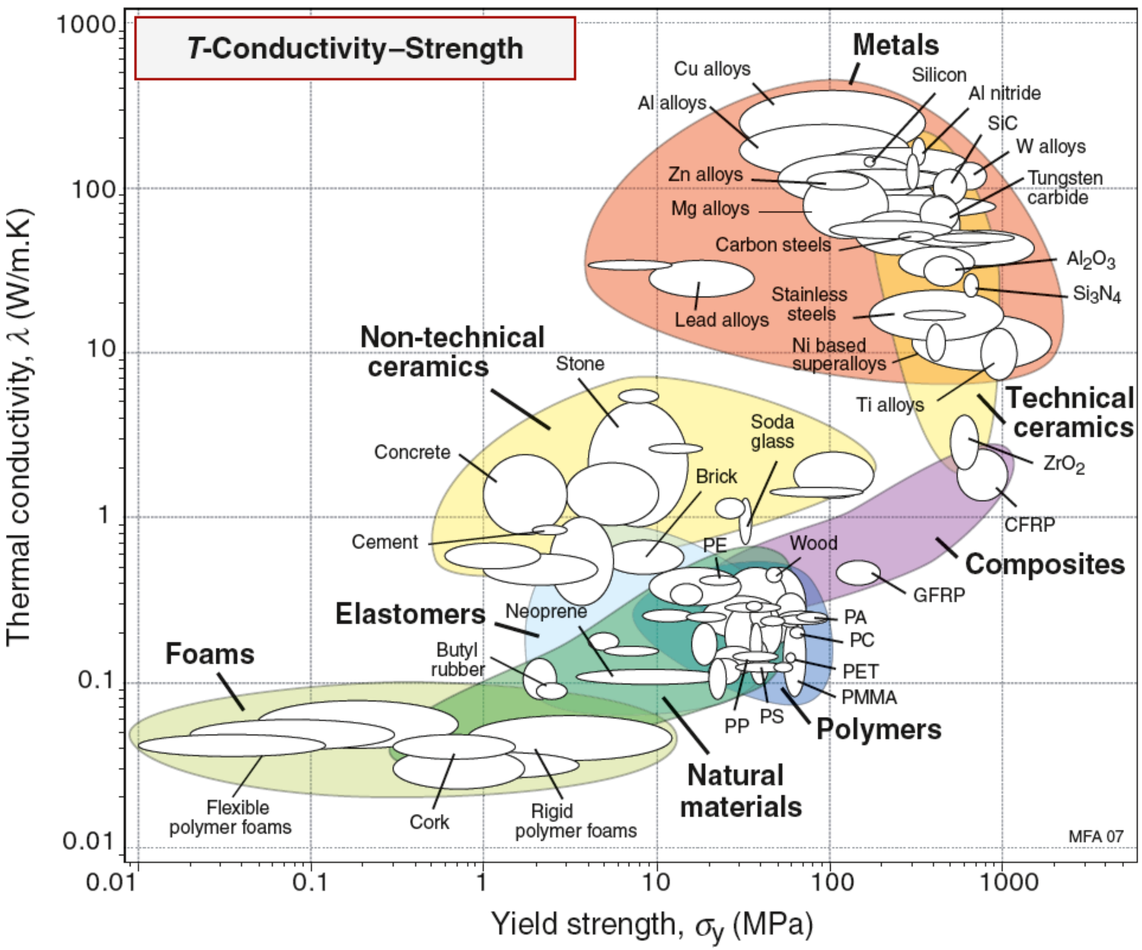


Figure 12.6

- Many applications require a combination of high strength and high conductivity
- Cu and Al are strong and good conductors, used in heat exchangers

Applications of thermal properties

- Thermal diffusivity

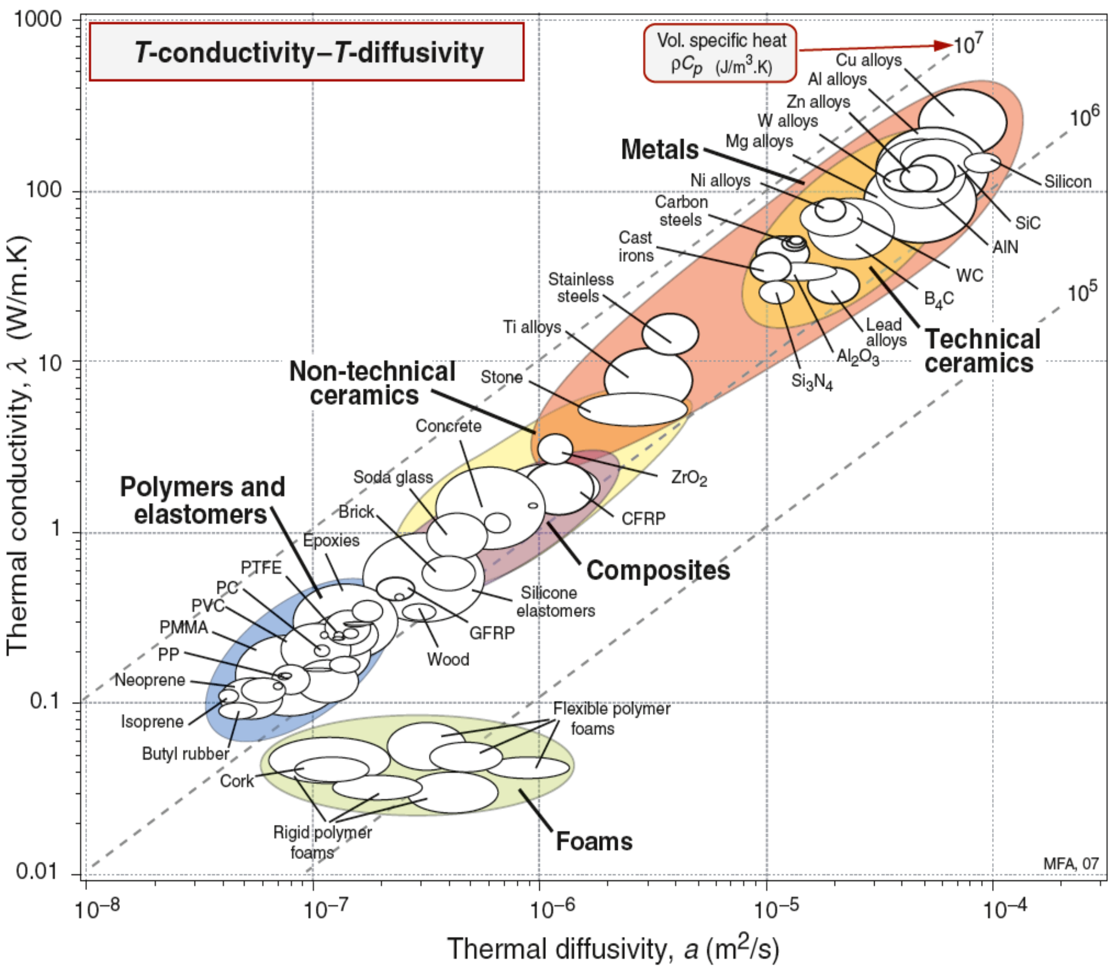


Figure 12.5

- Contours show the specific heat per unit volume
- Good conductors are located in the upper right
- Good insulators are found in the lower left
- E.g. foams are used for insulation because of their lower conductivities

Applications of thermal properties

- Thermal distortion

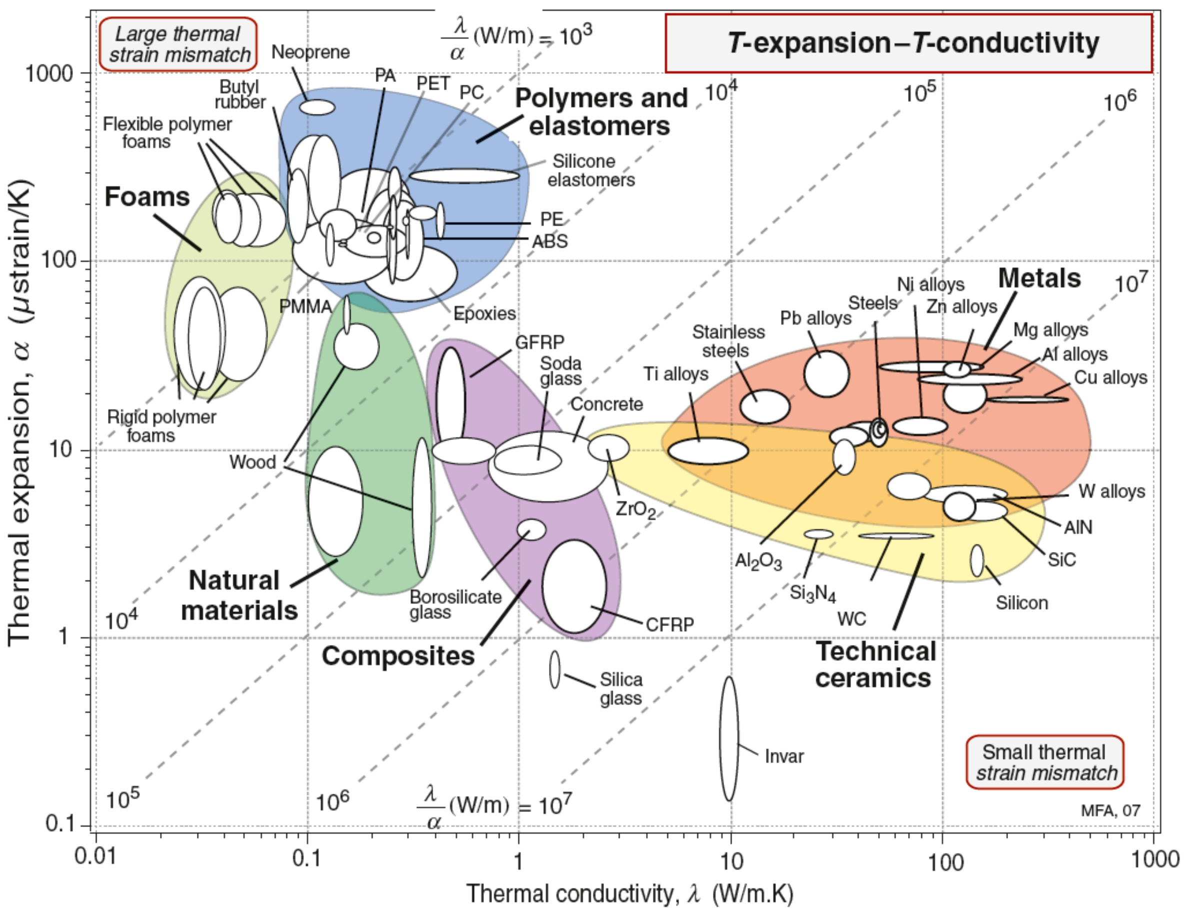


Figure 12.4

- Thermal distortion is a problem in fabrication
- Application of coatings is a typical issue, materials cool down at different rates
- Selecting materials that have similar thermal expansion coefficients from the chart

Summary

Thermo-mechanical behaviour:

- Coefficient of thermal expansion (CTE)

- Thermal stresses with temperature difference
- Thermal distortion
- Thermal shock

- Selection of materials for thermal applications

Learning outcomes

Thermal properties important in materials processing:

- Heat capacity
- Heat conduction
- Thermal expansion and contraction

Thermal properties important in materials applications:

- Thermal management – thermal diffusivity

- Thermal stresses
- Expansion
- Thermal cycling
- Thermal shock